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Roll No. .....

PAPER ID—21132

B.Tech. EXAMINATION, 2025

(Fourth Semester)

COMPUTER SCIENCE AND ENGINEERING

Computer Organization and Architecture

Time : 3 Hours *Maximum Marks : 70*

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Section. Q. No. **1** is compulsory. All questions carry equal marks.

1. (a) In what situations might fixed-point representation be preferable to floating-point representation ?

(b) Discuss the impact of memory hierarchy (cache, main memory) on bus traffic.

(c) How can logic microoperations be used to perform comparisons (greater than, less than, equal) ?

(d) What are status flags and how are they used by the ALU (Arithmetic Logic Unit) to indicate conditions (carry, zero) ?

(e) How does the control unit utilize the clock signal to synchronize operations within a processor ?

(f) What are the benefits of instruction pipeline ?

(g) What exactly is virtual memory, and why is it required ? Explain. 14

Section A

2. What do you mean by error detection code ? What are the types of error detection ? Explain. What are complements in computer architecture ? What is the purpose of Complements in Computer Architecture ? Explain with suitable example. What are the applications of complements ? 14

7. What is vector processing ? What are the characteristics of vector processing ? Discuss SISD SIMD MISD and MIMD with structural diagram of each 14

Section D

8. What are the IO device interfaces ? Explain with neat and clean diagram. Which are four types of commands received by an interface ? List out functions of Input-Output Interface. What is the need of IO interface in COA ? How does Data transfer between CPU and the I/O devices take place ? Explain. 14

9. What do you mean by associative memory ? Show the block representation of an Associative memory. What is mapping and explain associative and set associative mapping ? What is the difference between direct mapping and associative mapping ? Explain with example and neat and clean diagram. 14